

Question 18**1 mark**

Given $\theta = 240^\circ$, identify the coordinates of the point, $P(\theta)$, on the unit circle.

- a. $\left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$
- b. $\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$
- c. $\left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$
- d. $\left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$

Question 19**1 mark**

Given the functions, $f(x) = -3x + 5$ and $g(x) = x^2 + x - 1$, identify the value of $g(f(2))$.

- a. -10
- b. -5
- c. -3
- d. -1

Question 20**1 mark**

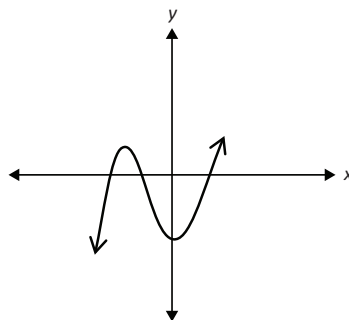
Given $f(x) = \sqrt{x} + 5$, identify the equation of the transformed graph that has the same y -intercept.

- a. $y = f(x - 3)$
- b. $y = -3f(x)$
- c. $y = f(-3x)$
- d. $y = f(x) - 3$

Question 21**1 mark**

Given the graph of $p(x) = ax^3 + bx^2 + cx + d$, identify the statement that is true.

- a. $a > 0, d < 0$
- b. $a > 0, d > 0$
- c. $a < 0, d < 0$
- d. $a < 0, d > 0$

**Question 22****1 mark**

Identify the general solution of the equation, $\csc\theta = -1$.

- a. $\theta = \frac{\pi}{2} + k\pi, k \in \mathbb{Z}$
- b. $\theta = \frac{3\pi}{2} + 2k\pi, k \in \mathbb{Z}$
- c. $\theta = \frac{3\pi}{2} + k\pi, k \in \mathbb{Z}$
- d. $\theta = \pi + 2k\pi, k \in \mathbb{Z}$

Question 23**1 mark**

Identify the equation of a radical function with a domain of $[-6, \infty)$ and a range of $(-\infty, 3]$.

a. $y = -\sqrt{x+6} - 3$

b. $y = -\sqrt{x+6} + 3$

c. $y = \sqrt{-(x+3)} - 6$

d. $y = \sqrt{x-6} + 3$

Question 24**1 mark**

Identify the value of n in the equation, ${}_nC_3 = {}_nC_7$.

a. 3

b. 4

c. 7

d. 10

Question 25**1 mark**

Identify the remainder when $p(x) = x^5 - 1$ is divided by $(x+1)$.

a. -2

b. -1

c. 0

d. 4

Question 26

2 marks

122

Match each function with its corresponding graph.

Write the corresponding letter in this column.

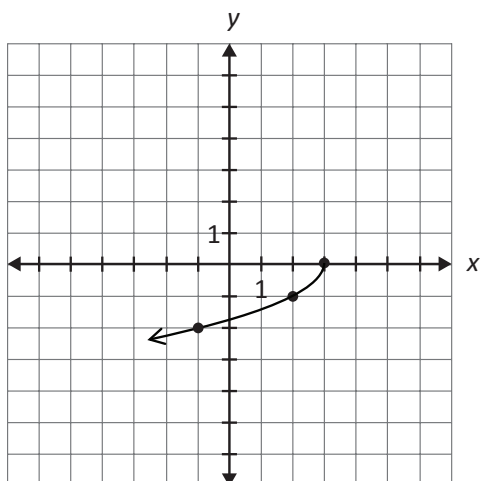
$$y = \sqrt{x} - 3 \quad \underline{\hspace{2cm}}$$

$$y = \sqrt{-x} + 3 \quad \underline{\hspace{2cm}}$$

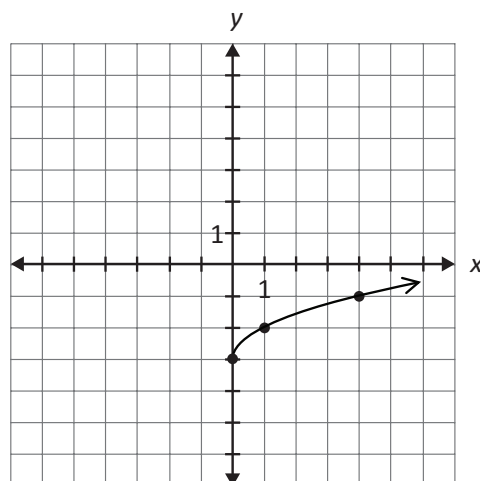
$$y = -\sqrt{x+3} \quad \underline{\hspace{2cm}}$$

$$y = -\sqrt{-(x-3)} \quad \underline{\hspace{2cm}}$$

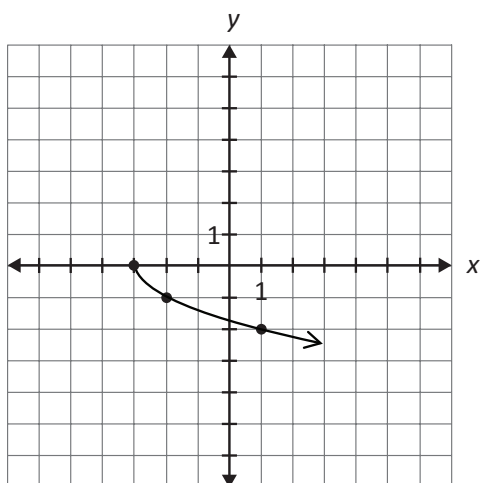
A)



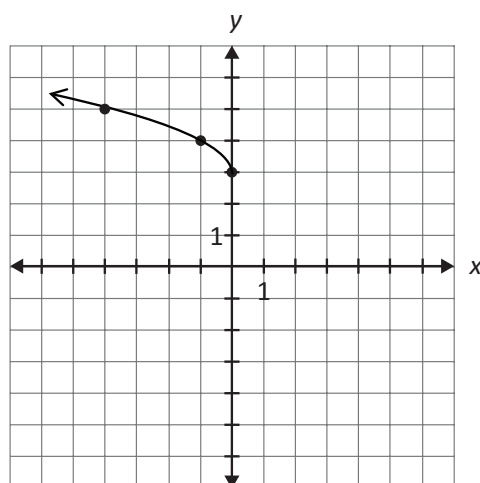
B)



C)



D)

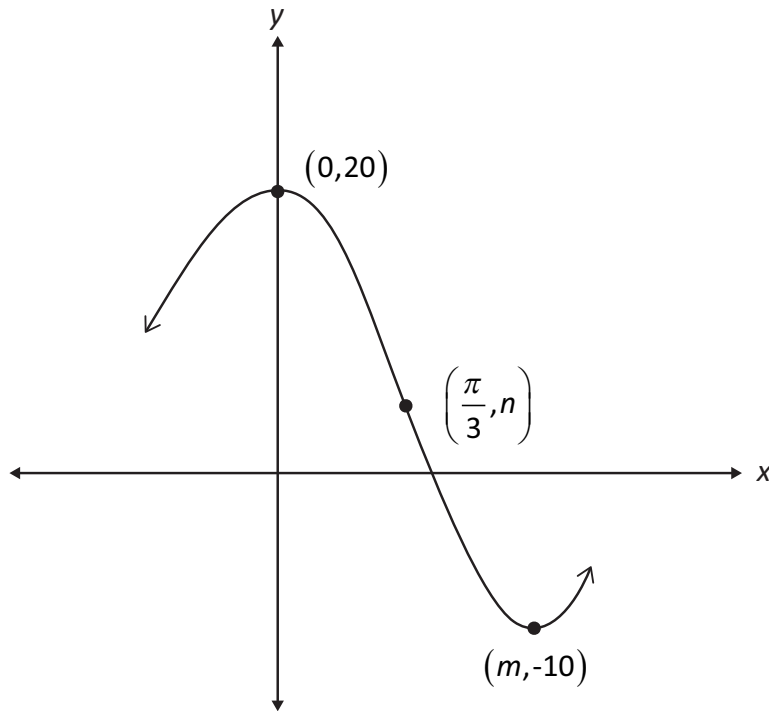


Question 27

2 marks

123

The amplitude, A , of the sinusoidal function can be determined using the equation $A = 20 - n$. State the values of m and n .



Question 28

3 marks

124

Solve, algebraically.

$${}_{n-1}P_2 = 20$$

Question 29

3 marks

125

Determine the exact value of $\sin\left(\frac{23\pi}{12}\right)$.

Question 30**1 mark**

126

Cameron was asked to solve the equation, $\log_3(x-4)=2$.

Cameron's solution:

$$x - 4 = 2^3$$

$$x - 4 = 8$$

$$x = 12$$

Describe Cameron's error.

Question 31

1 mark

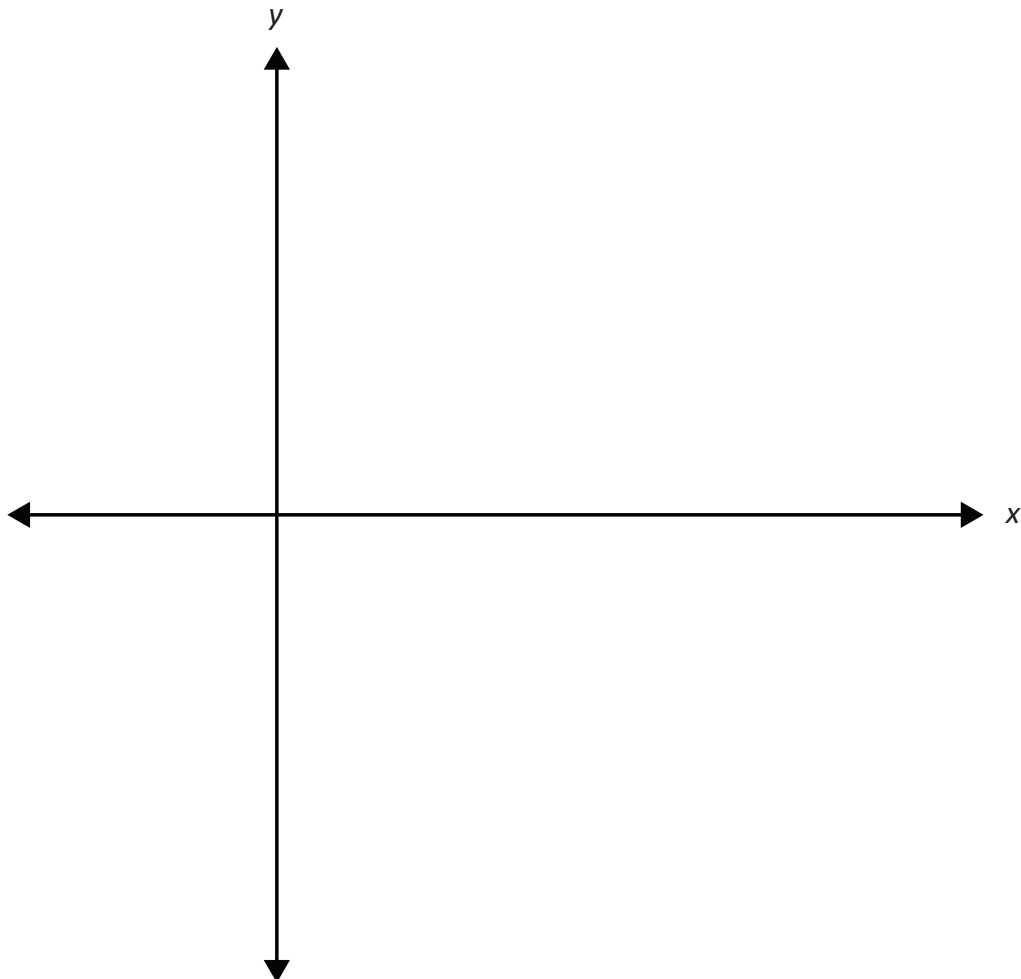
127

Given the identity $\sin^2 \theta = \frac{\tan \theta \sin \theta}{\sec \theta}$, state the non-permissible value of θ over the interval $[\pi, 2\pi]$.

Question 32**4 marks**

128

Sketch the graph of $y = -5\cos\left(\frac{\pi}{2}x\right) + 2$ over the domain $[0, 5]$.



Question 33

1 mark

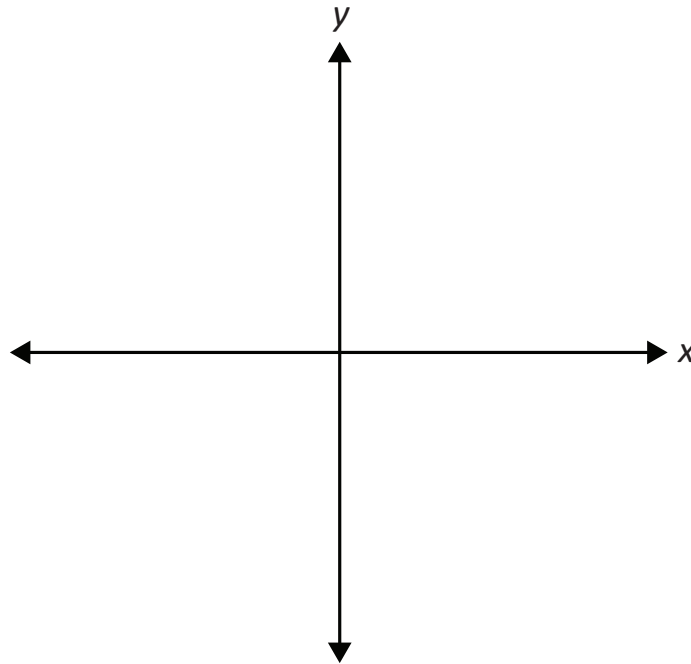
129

Describe the behaviour of the graph of the polynomial function, $p(x) = (x - 3)^2(x - 1)$, as it approaches the x-intercept at $x = 3$.

Question 34**1 mark**

130

Sketch the angle of 4 radians in standard position.



Question 35

4 marks

131

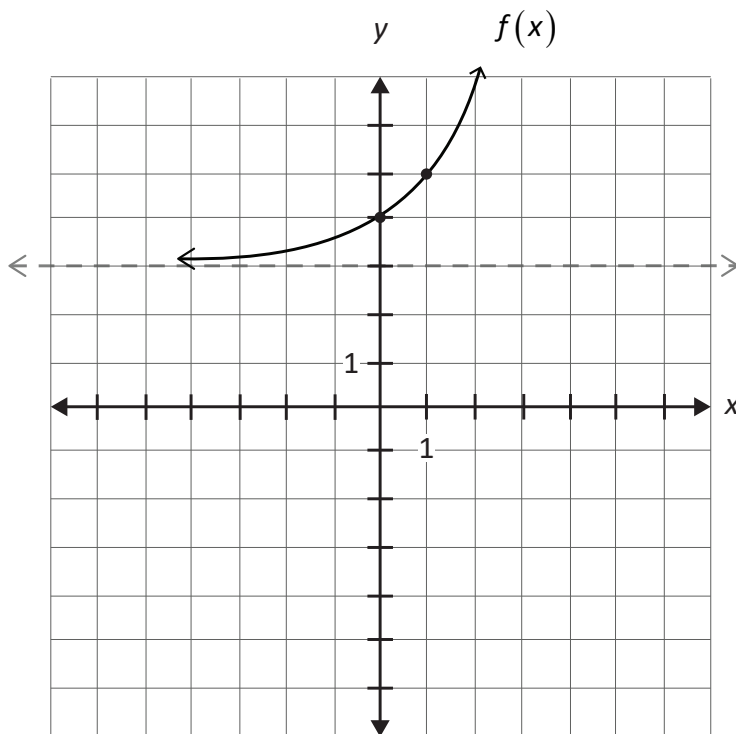
Solve, algebraically.

$$\log_5 12 + 2\log_5 x = \log_5 48$$

Question 36**2 marks**

132

State the equation of the exponential function represented by the graph of $f(x)$.



$f(x) =$ _____

Question 37**3 marks**

133

Evaluate.

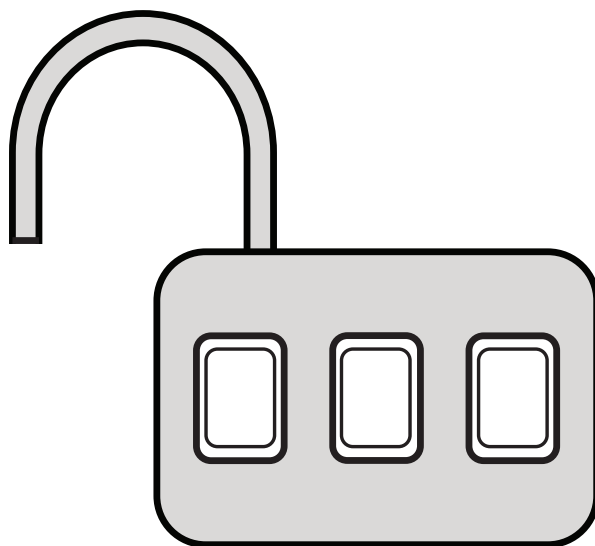
$$\cos\left(\frac{11\pi}{3}\right) + \csc\left(-\frac{\pi}{3}\right) \cot\left(\frac{11\pi}{6}\right)$$

Question 38

1 mark

134

Min Li has a 3-digit code for his lock. Determine the number of possible codes for his lock, if repetition is allowed.



Question 39

2 marks

135

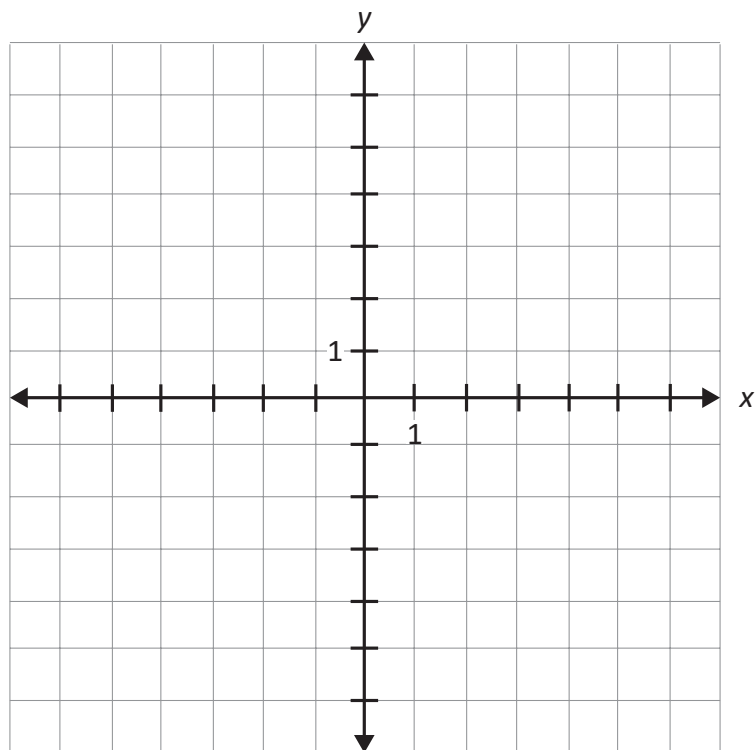
Solve, algebraically.

$$\left(\frac{1}{6}\right)^{3x+2} = 6^{2x}$$

Question 40**4 marks**

136

Sketch the graph of $f(x) = \frac{5}{(x-1)(x+3)}$ and state the y-intercept.



y-intercept: _____

Question 41**2 marks**

137

Given $\log_2 x = 5$, determine the value of $\log_4 (2x)$.

Question 42**1 mark**

138

Henry was asked to prove the identity, $\csc^2 \theta - 2\cos^2 \theta + 1 = \cot^2 \theta + 2\sin^2 \theta$, for all permissible values of θ .

His work:

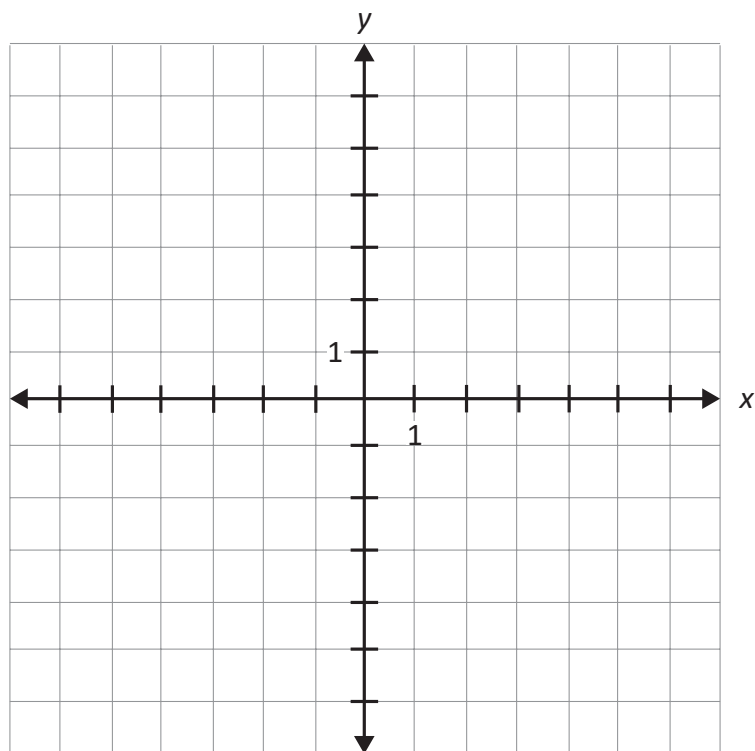
Left-Hand side	Right-Hand side
$\csc^2\left(\frac{\pi}{4}\right) - 2\cos^2\left(\frac{\pi}{4}\right) + 1$	$\cot^2\left(\frac{\pi}{4}\right) + 2\sin^2\left(\frac{\pi}{4}\right)$
$(\sqrt{2})^2 - 2\left(\frac{\sqrt{2}}{2}\right)^2 + 1$	$(1)^2 + 2\left(\frac{\sqrt{2}}{2}\right)^2$
$2 - 1 + 1$	$1 + 1$
2	2
LHS	RHS ✓

Explain why his proof is insufficient.

Question 43**4 marks**

139

Sketch the graph of $y - 2 = -\sqrt{x - 1}$.



Question 44**2 marks**

140

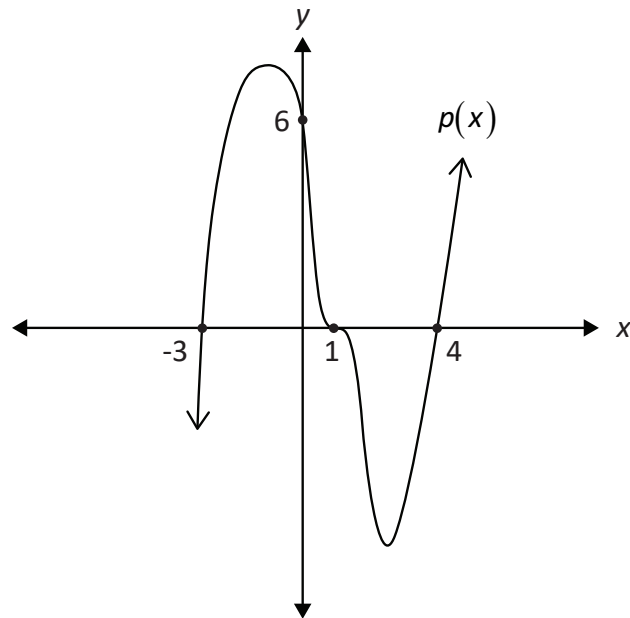
Evaluate.

$$\log_3 54 - \log_3 6$$

Question 45**2 marks**

141

Determine, algebraically, the equation of the graph of the polynomial function, $p(x)$.



$p(x) =$ _____

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